

Responsible AI Ethical, and Legal Aspects

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This lecture is not only about ethics, but ethics play a key role in responsible AI therefore it is important to provide a definition.



What is ethics?

“At its simplest, ethics is a system of moral principles. They affect how people make decisions and lead their lives.

Ethics is concerned with what is good for individuals and society and is also described as moral philosophy.”

Source: BBC Ethics Guide, "Introduction to Ethics," bbc.co.uk

What do ethics cover?

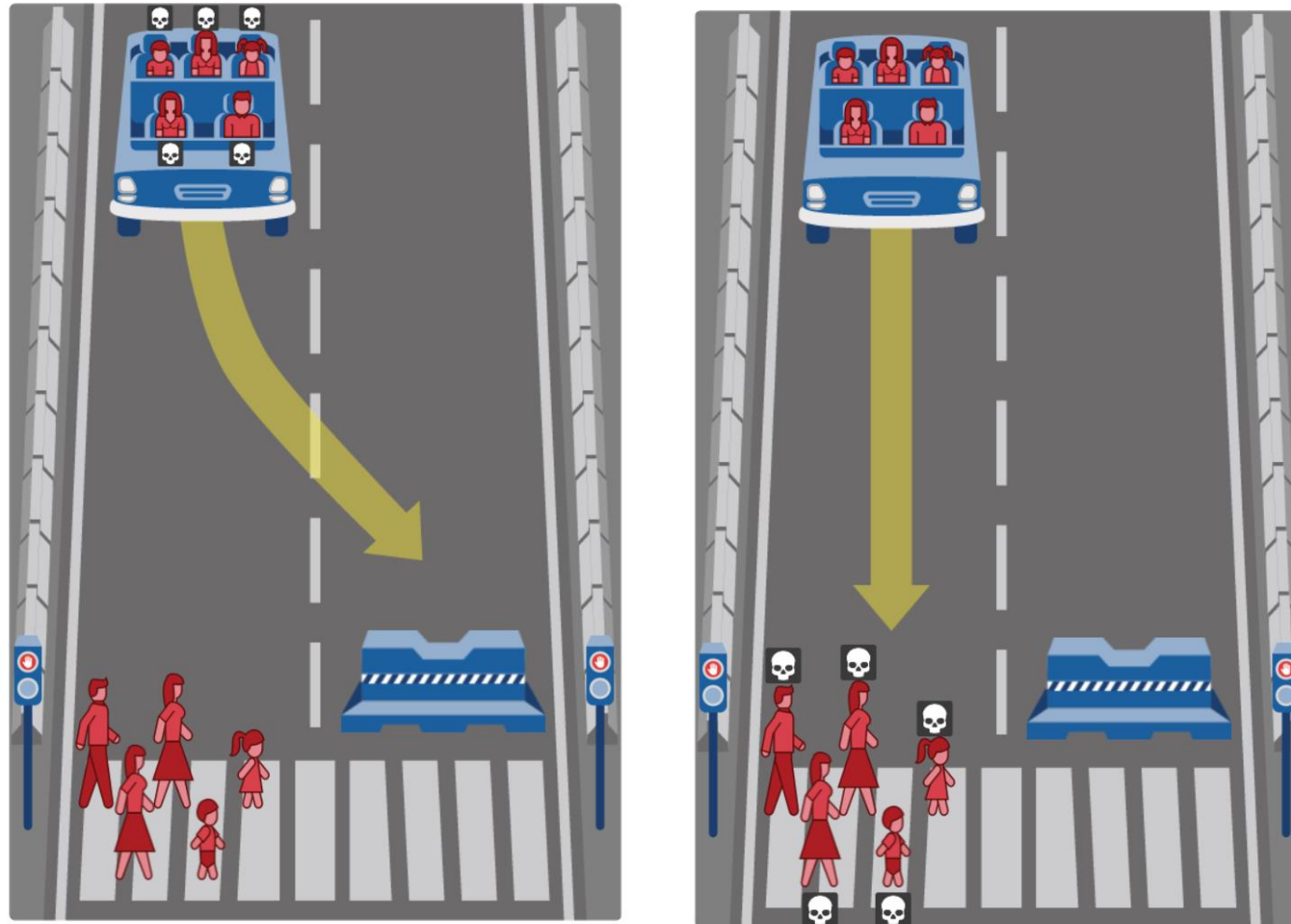
Ethics covers the following dilemmas:

- how to live a good life
- our rights and responsibilities
- the language of right and wrong
- moral decisions - what is good and bad?

Defining what is good or right is quite hard.

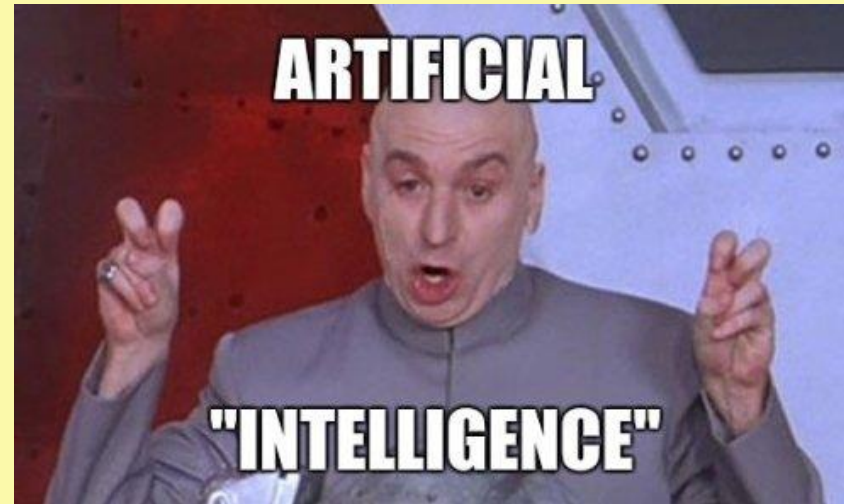
- Usually no easy answers.
- Many gray areas.
- Ethics change over time and is guided by the principles and values of people and society.
- Legal is not necessarily Ethical and vice versa.

Screenshot of a Moral Machine dilemma



By Iyad Rahwan - Own work, CC BY-SA 4.0,
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AI can be very risky



Examples on when AI can be very risky

1. A hiring AI rejects qualified candidates due to biased training data.
2. A facial recognition system misidentifies individuals, leading to wrongful arrests.
3. A self-driving car fails to recognize a pedestrian, causing a fatal accident.
4. Risk assessment systems for visa applications.
5. AI applications in medical devices, diagnosis, or patient monitoring.
6. Deploying an LLM for simple customer support queries can be highly unsustainable.
7. AI operates in a drone to kill enemy soldiers in battlefield

When is AI risky?

- AI applications can have potential consequences on individuals, groups, or society.
- Whenever a system impacts human lives, rights, or societal norms then it is a critical system.

Why is AI Dangerous and Riskier than a human making similar errors?

- **AI operates at scale**—small errors can have widespread impacts.
- **Lack of explainability** makes it hard to identify and address problems.
- It is **difficult to determine the Accountability**.
- **Perceived trustworthiness** due to its human-like behaviour.

Let us break down what can go wrong?

Unethical application of AI

- Bias in decision-making
- Unfair outcomes
- Unequal access to AI benefits.
- Unable to provide the right decision due to ethical dilemma
- Violating human rights

Let us break down what can go wrong?

Privacy and Data Protection

- Breach of sensitive information (what if your data leaked to an AI systems)
- Misuse of personal data (what if someone use your personal data in the training of the system) (public != publicized)

Let us break down what can go wrong?

Safety Risks

- Unpredictable outcomes in critical systems (e.g., healthcare, transportation).
- Vulnerability in AI applications can lead to disastrous outcomes

Let us break down what can go wrong?

Perceived trustworthiness of faked data

- **Disinformation and propaganda campaigns:** Public manipulation and elections
- **Financial Fraud:** in 2019, scammers used AI-generated audio to mimic a CEO's voice and tricked a UK-based energy firm into wiring \$243,000 to a fraudulent account.
- **Fake “Live” Support Agents:** Some companies use AI-generated “video agents” that look human but don't disclose they're synthetic.
- **Celebrity Pornography and Revenge Porn**



AI generated image posted by Donald Trump

Let us break down what can go wrong?

Environmental Impact

High energy consumption of AI models like GPT-4.

And Who is responsible when something goes wroing?

Accountability Challenges

Who is responsible when AI fails?

- Developer
- Company
- Society
- Regulators
- Deployers
- Users
-

Bottomline: Many stakeholder are involved, and it is important to define the roles and responsibilities

Use cases

Ethics, proxy bias, training data bias, data privacy, model error

Use-case: a model to predict retention rates of trainees, cadets or students

- ## Employers, School, Immigration offices



What is wrong with this application?

1. What kind of harm can be caused by the model?
2. How could the model cause harm?
3. Who is responsible for **the harm** and for the **model error**?
 - Researcher/developer?
 - Manager? Developing Company?
 - Deploying company?
 - Lawmaker?
 - Society?

Should we do this?

- Often, the "true label" we aim to predict is either unavailable or too complex to measure directly.
- This forces us to use a **proxy variable**—a measurable substitute that correlates with the true target.
- **Other examples:**
 - In healthcare, we might want to predict patient well-being and quality of life (true label), but we use overall survival (proxy) instead. Overall survived days are easier to measure but may not fully capture overall health.
 - A hiring algorithm optimizes for GPA (proxy) to predict job performance (true label). GPA alone might ignore interpersonal skills, creativity, or adaptability, which are critical for many roles.

- **True Label Unavailability or Complexity:**

Often the true label is unavailable / very complex.

- **Correlation vs. Causation:**

The proxy might only be correlated with the true target without a causal relation.

- **Proxy Insufficiency:**

Sometimes the proxy is not sufficient to reflect the optimal goal.

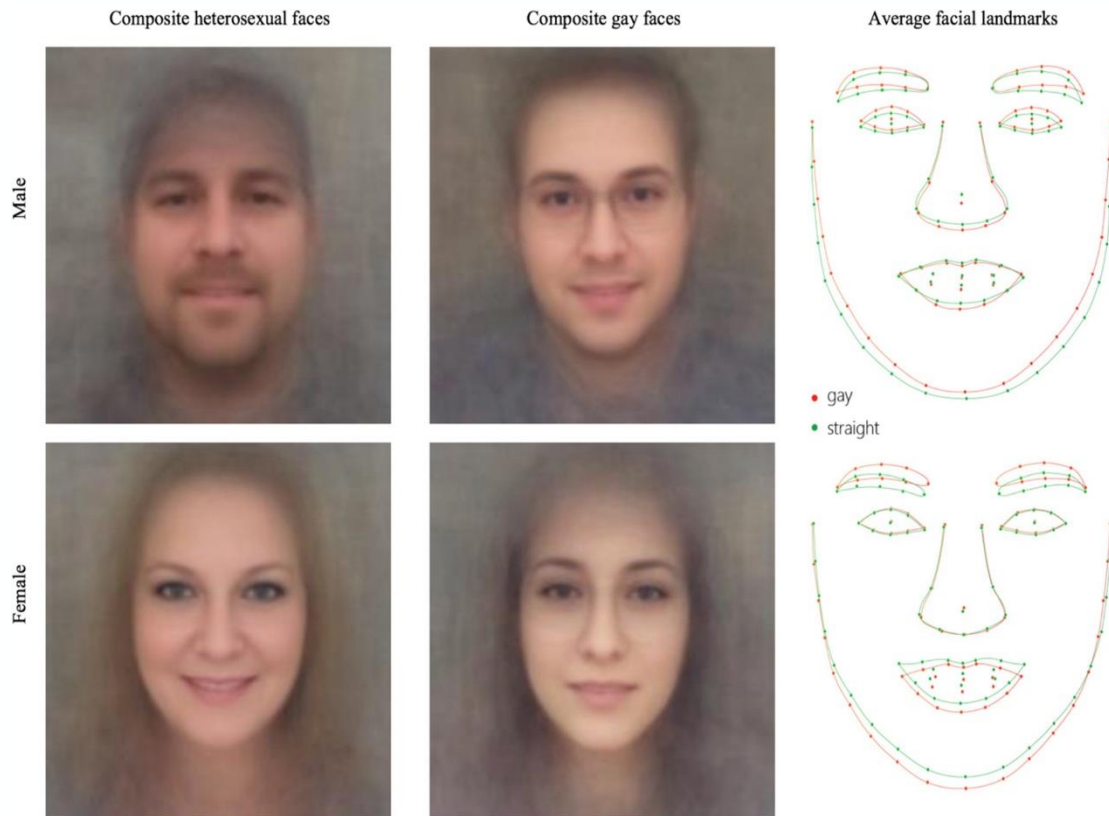
- **Proxy Bias:**

The proxy can be biased. For instance, measuring reoffending rate in the USA. Historically, there have been over-policed communities leading to higher reoffending rates)

Other examples

The "AI Gaydar"

Predict sexual orientation from facial images



Startup Faception

Facial Personality Analytics



Is the research question ethical?

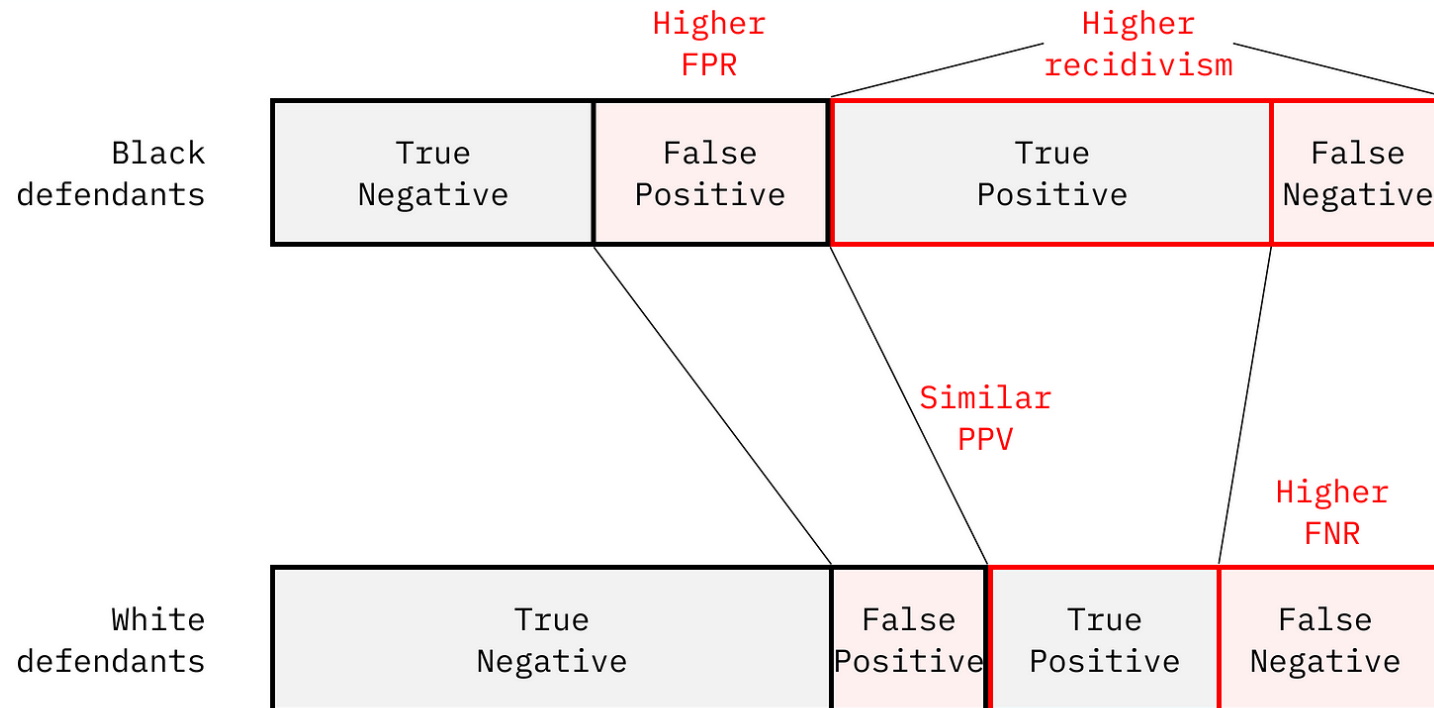
- Disclosing the sexual orientation can lead to discrimination. It can even lead to death penalty in some countries.
- It can affect a person's employment, family, relationships, educational or health care opportunities, ...
- Moreover, personal information like gender, race, sexual orientation, religion should be intimate and private.

Was it ethical to use the training data?

- Training data from a popular American dating website
 - 35,326 pictures of 14,776 people.
 - All white: Gay, straight, male and female represented evenly.
- Deep learning model to extract facial and grooming features and classifier to predict orientation.
- People did not intend their pictures to be used in other ways and did not agree to participate in the study
 - Legal $\neq$ Ethical
 - Public $\neq$ Publicized

Another Use-case with data bias

a model to predict the likelihood of a defendant reoffending (recidivism).



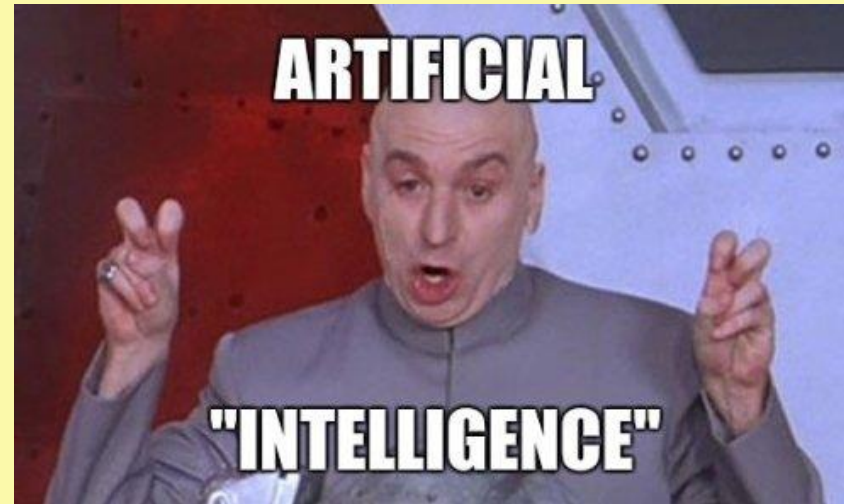
- Black defendants: More likely falsely labeled as high-risk.
- White defendants: More likely falsely labeled as low-risk.
- Historical data reflecting systemic bias let alone the biased proxy
- Balanced data does not necessarily represent true distribution

Improperly Built or insufficiently Audited AI Can lead to unfair outcomes

- Unfair AI can reinforce existing inequalities, leading to **ethical, social, and legal challenges**.
- **Fair AI ensures equitable treatment** by minimizing bias and preventing discrimination against any individual or group.
- It promotes **trust and accountability** in AI systems, especially in high-stakes areas like hiring, justice, and healthcare.

Who controls the training data, controls the output

Security risks

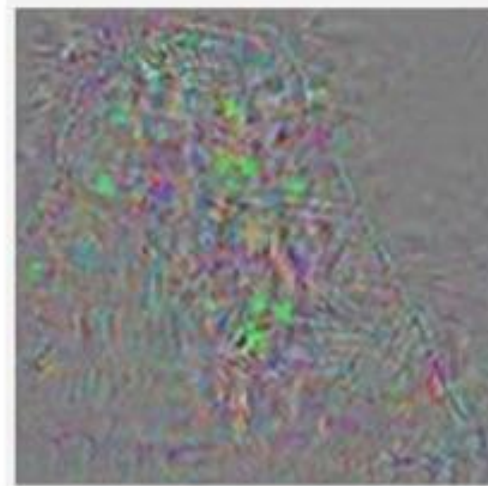


Attacking machine learning systems

Idea: Use gradient to compute small perturbation towards different class.



Original image

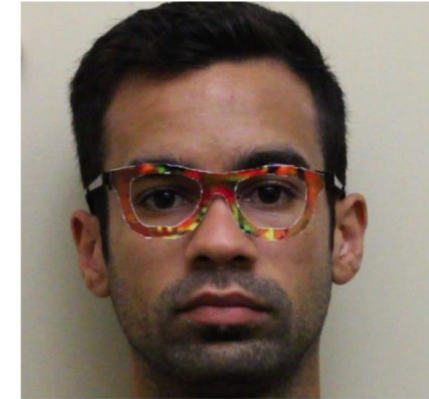


Perturbations



Adversarial example

Who knows the weights, can fool the model



"Accessorize to a Crime: Real and Stealthy Attacks on State-of-the-Art Face Recognition", Sharif, Bhagavatula, Bauer & Reiter.

Who knows the weights, can fool the model

AI EU ACT

It is now a law



AI EU Act

Objective

- *prevent potential risks of AI systems from arising by setting out regulations for their creation, deployment, and usage.*

'AI System' Definition

- *'a machine-based system that is designed to operate with varying levels of autonomy and that can, for explicit or implicit objectives, generate outputs such as predictions, recommendations, or decisions that influence physical or virtual environments.'*



AI EU Act risk-based approach

The AI EU Act moves the AI EU field to law enforcement



- > Prohibited AI Practices
- > Regulated high risk AI systems
- > Specific obligations
- > Permitted with no restrictions

EU AI Act categorization of critical AI applications

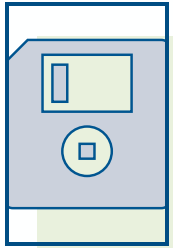
1. **Critical Infrastructure** (e.g., electricity grids, water supply).
2. **Education and Training** Example: Automated grading systems or recruitment tools.
3. **Employment and Workforce Management** AI systems used in hiring, firing, promotion, or performance evaluation.
4. **Access to Essential Services** Example: Loan approval algorithms or AI-driven diagnostic tools.
5. **Law Enforcement** Predictive policing or criminal risk assessment.
6. **Border Control and Migration** Example: Risk assessment systems for visa applications.
7. **Healthcare and Safety** :AI applications in medical devices, diagnosis, or patient monitoring.
8. **Product Safety**: AI embedded in products like vehicles, machinery, or toys.

AI EU Act risk-based approach

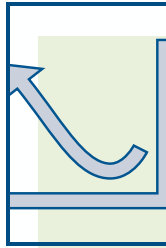
The Act establishes role-tailored obligations **depending on the level of risk from AI**



How to determine the risk category and the risk-benefit balance of any AI-based application?



What is the source of training data?



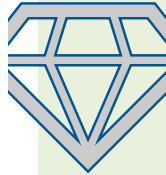
Who can modify training data and how?



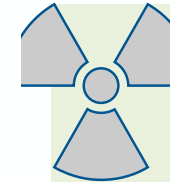
which potential damage may occur?



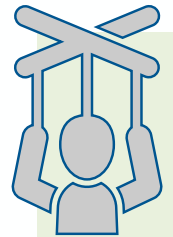
Direct impact on individual people?



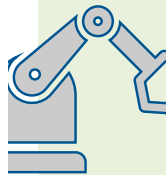
What is the business and financial impact?



What would be the reputational and financial risks?



How will the business deploy or leverage the AI solution?



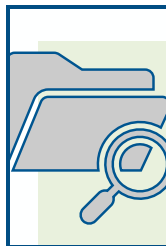
Autonomous outputs or judgment without human input?



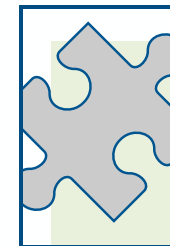
Did you build the AI system, buy, acquire, partner, subscribe?



Is this part of discovery process?



Are personal data are used as input to the AI model?



Is it to generate a hypothesis that will be later validated?

Needed for responsible and compliant AI

Ethical Guidelines

Align AI development with ethical principles like fairness, transparency, and non-discrimination.

Data Privacy

Ensure compliance with data protection laws (e.g., GDPR) and safeguard user data.

Bias Mitigation

Regularly test and minimize biases in data, algorithms, and outputs.

Robust Governance

Establish clear accountability, oversight, and documentation of AI systems.

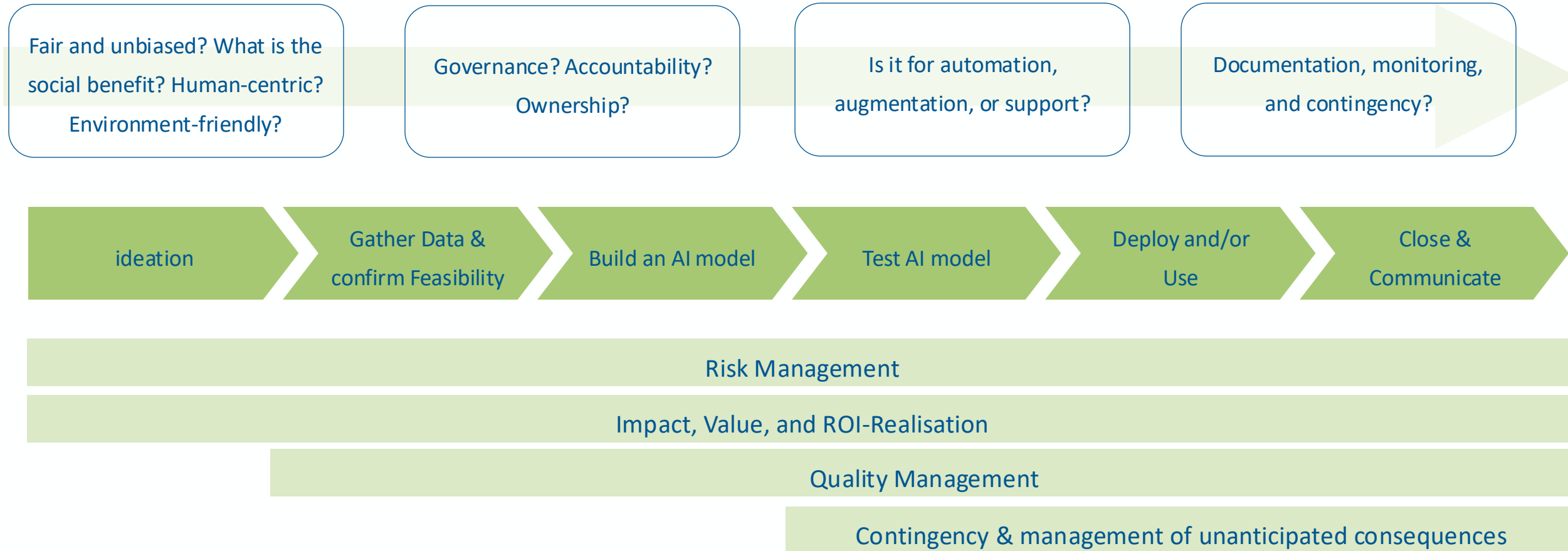
Security Measures

Protect AI systems against cyber threats and ensure system resilience.

On AI EU Act

'Provider' Definition	<i>Provider is defined as a natural or legal person that develops an AI system or that has an AI system developed and places that system on the market</i>
'Deployer' Definition	<i>Deployer is defined as any natural or legal person, under whose authority the system is used</i>
Prohibited AI Practices	<i>Some AI practices are outright banned, including using subliminal techniques, exploiting vulnerabilities of specific groups, creating a social score for individuals, and using real-time remote biometric identification systems in public spaces</i>
Definition reconciles two contrasting factors:	-retains sufficient adaptability to encompass prospective advancements in AI technology. -sets out a clear borders between AI systems and other advanced technology systems that do not qualify as AI
Scope	<i>The Act applies a horizontal approach, meaning it broadly covers all AI systems rather than targeting specific industries.</i>
No need of local implementation	<i>As a Regulation, its applies across all EU member states without the need for individual local transposition (like the GDPR).</i>

Responsible AI touchpoints across the AI lifecycle for developers and users



Q&A

